TRANSLATIONAL SCIENCE

Streptococcus species enriched in the oral cavity of patients with RA are a source of peptidoglycan-polysaccharide polymers that can induce arthritis in mice

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ABSTRACT

Objectives Analysis of oral dysbiosis in individuals sharing genetic and environmental risk factors with rheumatoid arthritis (RA) patients may illuminate how microbiota contribute to disease susceptibility. We studied the oral microbiota in a prospective cohort of patients with RA, first-degree relatives (FDR) and healthy controls (HC), then genomically and functionally characterised streptococcal species from each group to understand their potential contribution to RA development.

Methods After DNA extraction from tongue swabs, targeted 16S rRNA gene sequencing and statistical analysis, we defined a microbial dysbiosis score based on an operational taxonomic unit signature of disease. After selective culture from swabs, we identified streptococci by sequencing. We examined the ability of streptococcal cell walls (SCW) from isolates to induce cytokines from splenocytes and arthritis in ZAP-70-mutant SKG mice.

Results RA and FDR were more likely to have periodontitis symptoms. An oral microbial dysbiosis score discriminated RA and HC subjects and predicted similarity of FDR to RA. *Streptococcaceae* were major contributors to the score. We identified 10 out of 15 streptococcal isolates as *S. parasalivarius* sp. nov., a distinct sister species to *S. salivarius*. Tumour necrosis factor and interleukin 6 production in vitro differed in response to individual *S. parasalivarius* isolates, suggesting strain specific effects on innate immunity. Cytokine secretion was associated with the presence of proteins potentially involved in *S. parasalivarius* SCW synthesis. Systemic administration of SCW from RA and HC-associated *S. parasalivarius* strains induced similar chronic arthritis.

Conclusions Dysbiosis-associated periodontal inflammation and barrier dysfunction may permit arthritogenic insoluble pro-inflammatory pathogen-associated molecules, like SCW, to reach synovial tissue.

Key messages

What is already known about this subject?

- The prevalence of periodontitis is significantly increased in patients with rheumatoid arthritis (RA), and its severity is associated with RA severity.
- Periodontitis is driven by pathobiont periodontal bacteria derived from the oral microbiota. Susceptibility to periodontal infections with pathobiont strains, such as *Porphyromonas gingivalis*, is supported by streptococcal-rich plaques.
- Inflammatory arthritis in mice and rats can be induced by non-viable fragments of streptococcal cell walls (SCW), which is poorly degraded, pro-inflammatory and persistent.
- Microbial remnants have been identified in synovia of patients with RA and in rodent models of arthritis, induced after microbial infection.

What does this study add?

- RA and first-degree relatives were more likely to have periodontitis symptoms, and were discriminated from healthy controls (HC) by an oral microbial dysbiosis score.
- Members of the bacterial family *Streptococcaceae* were major contributors to the score.
- We propose a novel streptococcal species *S. parasalivarius* sp. nov., with strain-level genetic variation.
- SCW from RA and HC-associated *S. parasalivarius* strains induced similar mild chronic arthritis in mice.

How might this impact on clinical practice or future developments?

- Barrier dysfunction—associated with poor dental health, smoking and poor nutrition—may facilitate spread of insoluble microbial remnants to the synovium. Controlling mucosal inflammation and barrier dysfunction in genetically predisposed individuals may reduce dysbiosis and propensity for arthritis development.

INTRODUCTION

The aetiopathogenesis of rheumatoid arthritis (RA) interlinks female sex, age, genetic predisposition and environmental factors. Contributory environmental risks including smoking, particulate exposure, periodontitis and bronchiectasis implicate mucosal microbe/host immune interactions in disease development.[1 2] The gut and oral resident

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microbial communities, or microbiota, maintain homoeostasis by promoting immune cell function, tolerance and barrier functions. These include production of mucus, IgA and antimicrobial peptides.[3] Barrier protective mechanisms likely dictate the microbiota composition by supporting the environment of commensals in health or disrupting barrier function with pathobiont outgrowth in disease. Shifts in microbiota, termed dysbiosis, are associated with disease susceptibility. Dysbiosis of faecal and oral microbiota, and increased abundance of oral-resident bacteria in the faecal microbiome in patients with RA suggest that differential abundance of particular taxonomic groups may contribute to RA development or control.[4–8] Pathogen-associated molecular patterns (PAMPs) associated with pathobionts, which thrive or outgrow in dysbiotic mucosal environments, could activate the innate immune system in genetically-susceptible individuals, such as patients with or at risk of RA.[3] Microbial DNA remnants identifiable in the synovial fluid, membranes and serum of patients with RA may contribute PAMPs, as demonstrated in experimental models of arthritis induced after microbial infection.[9–11] However, the contribution of such potentially inflammatory microbial products to disease development or perpetuation is not fully understood.

The prevalence of periodontitis is significantly increased in patients with RA, and its severity is associated with RA severity.[12] Periodontitis is driven by pathobiont periodontal bacteria derived from the oral microbiota, including *Porphyromonas gingivalis* and *Aggregatibacter actinomycetemcomitans*.[13] *P. gingivalis* may function as a 'keystone pathogen', central to the transition to a dysbiotic community by promoting growth of and communicating with accessory bacteria, such as *Streptococcus gordonii*.[14] *A. actinomycetemcomitans* may also function as a keystone pathogen, where its combination with *Filifactor alocis* and *Streptococcus parasanguinis* is associated with localised aggressive periodontitis.[15] While *P. gingivalis* may contribute to the aetiology of RA,[16 17] the evidence is still associative. For example, *P. gingivalis* antibody titres correlated with RA disease activity,[18] treatment of periodontitis reduced RA severity and disease-modifying anti-rheumatic drugs (DMARDs) simultaneously reduced RA disease activity, *P. gingivalis* antibody titres and periodontitis.[18 19] However, a prospective cohort failed to show that subjects with periodontal disease had an increased risk of later RA.[20] *P. gingivalis* cell-associated DNA was more likely to be detected in synovial fluid of patients with RA than in healthy controls.[21] These data suggest that while not causative, periodontitis-derived bacteria or debris reaching the blood may enhance inflammatory responses and autoantibody generation by spreading to synovial inflammatory tissue.[22]

Streptococci are a major group of early colonisers of dental plaque, making up about 80% of early biofilm constituents.[23 24] *S. sanguinis* is a pioneer organism that adheres to the tooth enamel through specific surface adhesins, allowing later more virulent colonisers, such as *S. mutans* to build a stratified, complex biofilm.[23] Such biofilms, coupled with poor oral hygiene, permit expansion of opportunistic pathogenic bacteria, thus increasing susceptibility to periodontal infections like *P. gingivalis*, supported by streptococcal-rich plaques.[25] Increased and decreased abundance of *Streptococcus* spp have been reported within the oral microbiome of patients with RA.[26–28] *Streptococcus* was observed at increased abundance in the gut of patients with RA and genetically at-risk individuals prior to disease onset.[29–31] Non-viable fragments of streptococcal cell wall (SCW) peptidoglycan-polysaccharide (PG-PS) induce inflammatory arthritis in certain strains of mice and rats after systemic administration. Notably, PG-PS is poorly degraded, pro-inflammatory and persistent in tissues.[32–34] Therefore, further exploration of the abundance of streptococci in RA is of clinical interest both for their potential role in the development of periodontitis and as a source of inflammation.

The tongue has the most stable, diverse and dense bacterial biofilm of the oral cavity.[35] To evaluate oral dysbiosis of the tongue biofilm and potential contributors supporting periodontal disease in patients with RA and genetically and environmentally at-risk individuals, we studied oral microbiota derived from tongue swabs in patients with RA, first-degree relatives (FDR) of RA probands and individuals without RA matched for age and sex (healthy controls, HC). We then employed a susceptible mouse strain to discern how systemic spread of pathobiont-derived PAMPs might contribute to inflammatory arthritis.

MATERIALS AND METHODS

Cohort and oral swab collection

In a study approved by the University of Queensland, and Gallipoli Human Research Ethics Committees, 116 patients with RA, 63 FDR and 43 HC, predominantly from an Australian urban environment, provided informed consent and clinical, lifestyle and sociodemographic data. Patients with RA and FDR were recruited in the waiting rooms of metropolitan rheumatologists, through community advertising and word of mouth (friends and neighbours). Each individual came to a clinical appointment involving history, questionnaire (online supplemental table 1) and a 68 tender and swollen joint count. Adult patients with RA were included if they either met American College of Rheumatology (ACR) 2010 classification criteria or had a confirmed diagnosis by a rheumatologist. FDR included parents, full siblings or offspring of patients with RA and were clinically ascertained as non-RA. HC were community volunteers without a family history of RA, but sometimes had other diseases. Two oral swabs taken from the centre of the tongue of each participant were stored at −80°C within 3 hours.

SCW arthritis induction in SKG mice

Female ZAP70^{W163C} BALB/c (SKG) mice aged 7 to 16 weeks were maintained under specific pathogen free conditions at Translational Research Institute. The University of Queensland animal ethics committee approved all experiments. Female SKG mice (n=6 per group) were injected intraperitoneally two times (days 0 and 22) with either 15 µg/gr body weight 10S PG-PS or SCW from axenic cultures, no treatment control or once with 3 mg curdlan. Joints were scored visually by an independent technician.[36] Decalcified ankle joints were stained with H&E for histological scoring.[37]

Additional methods for DNA extraction, amplicon sequencing, sequence processing and statistical analysis, axenic cultures, *Streptococcus* genus verification and characterisation, SCW isolation and responses in vitro, are provided in the online supplemental material.

RESULTS

The oral (tongue) microbiome differs in patients with RA and healthy controls

Two hundred and twenty-two individuals, comprising 116 treated, established patients with RA, 63 FDR and 43 HC subjects (table 1) were included. FDR all had a family history of RA. Of the control subjects, 24% had a relative with RA but none was a FDR. While the age of patients with RA and control subjects did not differ, FDR were significantly younger. Sixty-five per cent of patients with RA and 4% of FDR were anti-citrullinated peptide

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Table 1 Demographic features of the cohort

Feature	RA (n=116)	FDR (n=63)	HC (n=43)	P value
Age (mean, SD)	53.5 (14)	45.7 (16)	51.2 (15)	0.003
RA duration (median, range)	12 (1 to 46)	0	0	
Female sex (%)	92 (79.3)	42 (66.7)	31 (72.1)	0.169
Family history RA (%)	60 (51.7)	63 (100)	11 (25.5)	**0.015**
Alcohol consumer (%)	75 (64.7)	53 (84.1)	36 (83.7)	**0.003**
Smoking: never (%)	61 (52.6)	43 (68.3)	27 (62.8)	**0.018**
Ex	46 (39.7)	11 (17.5)	15 (34.9)	
Current	8 (6.9)	8 (12.7)	1 (2.3)	
Treatment, cs/b DMARDs (%)	96 (82.8)	0 (0)	2 (4.7)	**<0.0001**
Treatment, prednisone (%)	31 (26.7)	0 (0)	0 (0)	**<0.0001**
Bleeding gums last 5 years (%)	28 (24.1)	18 (28.6)	4 (9.3)	**0.055**
Tooth or gum infection in last 2 years	15 (12.9)	6 (9.5)	2 (4.7)	0.304
Streptococcal or throat infection last 2 years (%)	20 (17.2)	8 (12.7)	4 (9.3)	0.404
Chest infection requiring antibiotics last 2 years (%)	27 (23.3)	16 (25.4)	6 (14.0)	0.341
RF^+ (%)	70 (61.9)	6 (9.5)	2 (4.7)	**<0.0001**
$ACPA^+$ (%)	61 (54)	2 (3)	0 (0)	**<0.0001**
Tender joint count, of 68 (median, range)	0 (0 to 46)	0 (0 to 2)	0 (0)	**<0.0001**
Swollen joint count, of 68 (median, range)	0 (0 to 34)	0 (0 to 1)	0 (0)	**<0.0001**

$ACPA^+$, anti-citrullinated peptide autoantibody; cs/b, conventional synthetic/biological; DMARDs, disease-modifying anti-rheumatic drugs; FDR, first-degree relatives; HC, healthy controls; RA, rheumatoid arthritis; RF^+, rheumatoid factor.

autoantibody $(ACPA)^+$, respectively, while 56% of patients with RA and 12% of FDR were rheumatoid factor $(RF)^+$. Patients with RA were significantly less likely to drink alcohol or to have never smoked. Patients with RA and FDR tended to be more likely to have a history of bleeding gums—a symptom of periodontitis—than HC (p=0.055). Other infections were not more frequent in patients with RA.

After targeted 16S rRNA gene amplicon sequencing of one oral (tongue) swab from each individual, in two batches, preliminary principal component analysis (PCA) and one-way analysis of variance based on the first three components from PCA confirmed the absence of batch effect. We identified a microbial signature discriminating between patients with RA and HC using sPLS-DA (figure 1A) which included representatives of several bacterial families including *Enterobacteriaceae*, *Peptostreptococcaceae*, *Veilonellaceae* and *Streptococcaceae*. The importance and contribution of each operational taxonomic unit (OTU) defining the sPLS-DA component is represented in figure 1B.

The OTUs constituting this signature are combined in a sPLS-DA component and represent a 'microbial dysbiosis score' for each subject. A receiver operating characteristic curve constructed based on the dysbiosis score (AUC (area under

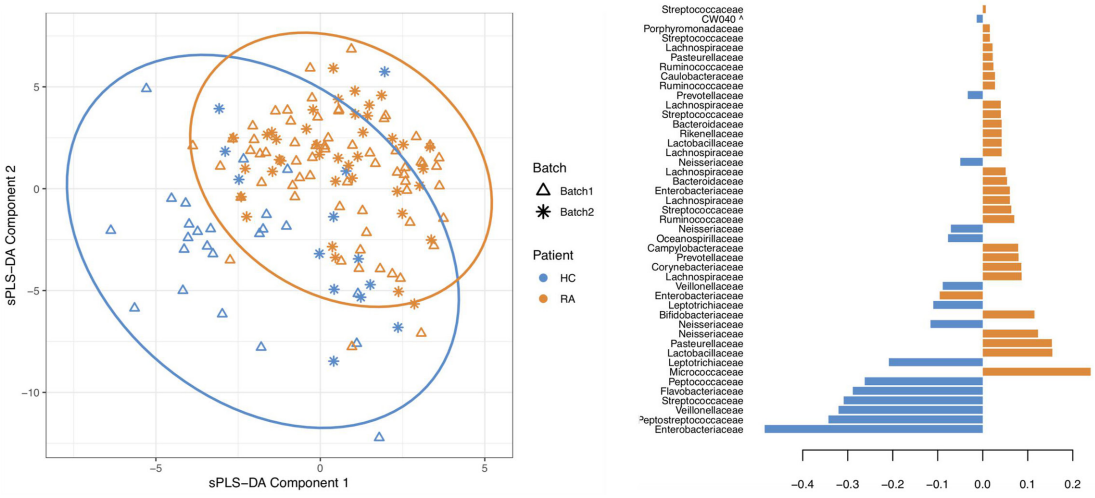


Figure 1 sPLS-DA sample plot of oral microbiome and OTU selection and contribution comparing patients with RA and healthy controls. (A) sPLS-DA sample plot of oral microbiome. The overall abundance of oral microbiota based on 16S rRNA gene amplicon sequencing is compared in HC subjects (HC, blue) and patients with RA (RA, orange). Samples were run in two sequencing batches, represented by open triangle and star symbols. The sPLS-DA model identified 45 OTUs on each component, which in combination help to discriminate the sample groups. 95% confidence ellipse plots are represented within each sample group. (B) Selected OTUs and their contribution in the sPLS-DA model when comparing patients with RA and HC. To reflect the importance of each OTU selected in sPLS-DA component 1, their loading weight coefficient is represented on the x-axis. A high (absolute) value indicate a highly discriminant OTU (bottom of the plot). Colours indicate the sample group (RA orange, HC blue) for which a given OTU is most abundant when considering their median value. HC, healthy controls; OTU, operational taxonomicunit; RA, rheumatoid arthritis.

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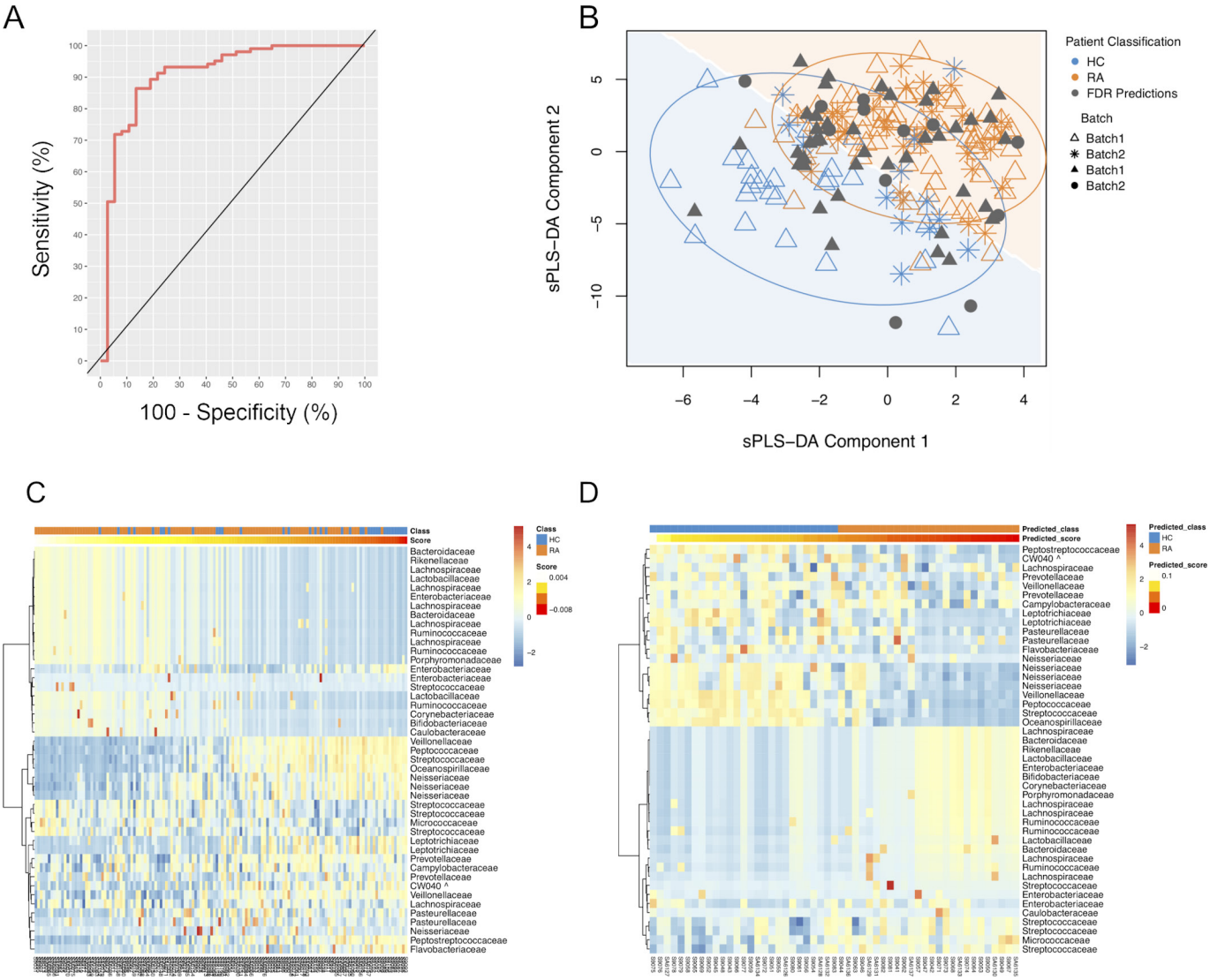


Figure 2 Oral microbiome signature to predict RA and classification of FDR according to the signature (A) Receiver operating characteristic curve (red) illustrates the ability of sPLS-DA to discriminate HC versus RA based on the identified microbial signature, with an AUC=0.9079. (B) The FDR samples are overlaid onto the sample plot from figure 1A for group prediction based on the sPLS-DA model fitted on the HC versus RA samples and the identified microbial signature (figure 1B). The background colour indicates the prediction of the two batches of FDR sample sequencing are denoted by grey circle and triangle symbols. (C) Heatmap representation of the oral microbiome signature from RA and HC samples. Abundance ranges from low (blue) to high (red) for each OTU (rows, Family or Order taxonomy) and each sample (columns). In rows, OTU are clustered according to Euclidean distance and Ward linkage method, while in columns the samples are ordered according to their microbial score (as indicated at the top of the graphic, second row) to highlight patterns of microbial abundance from the sPLS-DA. To illustrate the abundance patterns of the microbial signature identified by sPLS-DA and to summarise as a continuous disease score, the first row indicates the known group status, and the second row the proposed dysbiosis score. (D) Heatmap representation of the OTU signature for the FDR samples calculated from the sPLS-DA model and signature identified from the RA versus HC. In rows, OTU are clustered according to Euclidean distance and Ward linkage method while in columns the samples are ordered according to their predicted microbial score. The first row indicates the predicted status (as disease class is unknown), and the second row the proposed dysbiosis score showing a continuum ranging from RA-like (left columns) to HC-like score (right columns). AUC, area under the curve; FDR, first-degree relatives; HC, healthy controls; OTU,operational taxonomic unit; RA, rheumatoid arthritis.

the curve)=0.9079) discriminated non-RA from RA samples (figure 2A). The dysbiosis score was then used to predict which FDR were most or least 'RA-like' (figure 2B) based on the model fitted on the HC versus RA samples (figure 1). The oral microbiome signature (figure 2C and D) displays a heat map of abundance for each OTU, ranked according to the dysbiosis score, along with group status (for RA and HC) or predicted RA or non-RA status (for FDR) (online supplemental table 4). These data indicate that oral community profiles can be distinguished in patients with RA, which we suggest represents a dysbiotic state. Using sPLS-DA as a prediction tool, the oral microbiome of FDRs can be classified, or assigned to a spectrum, based on similarity to the RA oral dysbiosis score. However, we found no association between measured clinical parameters of the RA and HC samples and the dysbiosis score (online supplemental table 5).

Since Streptococcaceae were major contributors to oral dysbiosis in patients with RA and RA-like FDR, and their cell walls promote immune activation and arthritis in susceptible rodent strains, we isolated axenic streptococcal cultures from the

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remaining stored tongue swab of three patients with RA, three FDRs and three HC, to recover 15 isolates. Each isolate was identified based on morphology (online supplemental figure 1), 16S rRNA gene sequence similarity to existing strains and subsequent whole genome sequencing (table 2). Clinical features and dysbiosis scores of the subjects are also indicated (table 2). The majority (n=10) of the isolated strains, including all RA-associated isolates, were *S. sp001556435*, defined by the Genome Taxonomy Database as a distinct sister species to *S. salivarius*[38] and for which we propose the name *S. parasalivarius* sp nov. Other recovered isolates were either *S. salivarius* (n=1) or *S. parasanguinis_B* (n=4). Comparison of the recovered genomes to the *Streptococcus* OTUs within the discriminatory OTU profile (online supplemental table 4) suggested that two of the five OTUs likely belong to either *S. parasalivarius* or *S. salivarius* (100% identity to both species) and one OTU to *S. parasanguinis_B* (online supplemental table 6). The remaining two discriminatory streptococcal OTUs had highest matches to two unrecovered streptococci, *S. sp000187445* and *S. pneumoniae*, with the latter being a low-confidence match (94.4% identity).

To determine whether streptococcal strains of the same species isolated from patients with RA and HC differed at the genomic level—potentially translating to strain functional differences and inflammatory properties—we compared the genomes of the *S. parasalivarius* isolates RA: 21.1, 22.1 and 23.2, and HC: 2.1 (table 2, green shading). Phylogenetic comparison of these, and *S. parasalivarius* and *S. salivarius* isolates recovered from FDRs, clearly distinguished each person's strain (figure 3). Where multiple isolates were obtained from the same individual, all were closely related, for example, 2.1 and 3.1 (figure 3). SNP and indel analysis in comparison to the *S. parasalivarius* reference isolate CCH5-D3 identified over 250 RA-associated mutations (conserved among isolates from patients with RA) classified as moderate or high risk to protein structure, for example, missense, non-sense and frameshift mutations (online supplemental table 7). At the protein level, comparison of homologous proteins identified the three RA-associated isolates encode the galactose/lactose operon described in *S. salivarius*,[39] and a multidrug resistance efflux transporter (EmrE superfamily) absent from the *S. parasalivarius* isolate obtained from a HC (online supplemental table 8).

Since the genomic diversity observed between the RA-associated strains might impact functional differentiation, we compared the inflammatory potential of *S. parasalivarius* isolates 21.1, 22.1 and 23.2. We incubated heat-killed cells of these RA-associated isolates with splenocytes from naïve SKG mice, which are genetically susceptible hosts for Th17-dependent autoimmune arthritis in response to microbial triggers.[40] Two additional *S. parasalivarius* isolates, (HC isolate 2.1 and FDR isolate 16.1) and one *S. salivarius* isolate (FDR isolate 11.1) were assayed for reference. While in vitro production of tumour necrosis factor (TNF) and interleukin (IL)-6 differed between the RA-associated *S. parasalivarius* strains, with 21.1 and 22.1 inducing significantly lower levels of both cytokines than 23.2, TNF was the dominant cytokine induced by all isolates. TNF was produced at 10-fold higher levels than IL-6 and 100-fold higher levels than IL-10 (figure 4A–C). We compared SNPs and indels of each strain to ascertain genetic differences that might account for fine differences in TNF production between strains. Over 340 SNPs and indels classified as moderate or high risk were shared between strains 21.1 and 22.1 while absent from other assayed *S. parasalivarius* isolates (online supplemental table 9). Conversely, 171 SNPs and indels were identified as absent from 21.1 and 22.1 while present in the other strains (online

Table 2 Streptococcal isolate genome characteristics and clinical features of subjects

NCBI assembly accession	Isolate	16S rRNA gene-based identification	Genome-based classification	Genome completeness	Genome contamination	Genome contigs	Genome length	Source	Age	Sex	BMI	Infection history	RA drugs	ACPA titre	RF (IU)	Dysbiosis score x10^{-3} (rank)
JACLQX000000000	1.1	*parasanguinis*	*parasanguinis_B*	100	0.07	50	2 103 580	HC				Tooth	–	0	0	−1.43 (96)
JACLQW000000000	2.1	*salivarius*	*parasalivarius*	99.38	0.66	62	2 397 206	HC	35	F	26					
JACLQV000000000	3.1	*salivarius*	*parasalivarius*	99.38	0.66	53	2 395 186	HC								
JACLQU000000000	6.1	*parasanguinis*	*parasanguinis_B*	100	0.35	28	2 135 306	HC	54	M	31	Tooth	–	0	0	−2.26 (104)
JACLQT000000000	7.1	*parasanguinis*	*parasanguinis_B*	100	0.35	30	2 138 533	HC				Throat, sinus				
JACLQS000000000	9.1	*salivarius*	*parasalivarius*	99.76	0.15	39	2 127 501	FDR	22	F	24	Tooth	–	0	0	28.72 (19)
JACLQR000000000	10.1	*salivarius*	*parasalivarius*	99.76	0.15	42	2 128 725	FDR								
JACLQQ000000000	11.1	*salivarius*	*salivarius*	99.84	1.12	35	2 198 364	FDR	67	F	24	Tooth	–	0	0	7.94 (53)
JACLQP000000000	12.1	*parasanguinis*	*parasanguinis_B*	100	0	50	2 088 454	FDR				UTI				
JACLQO000000000	15.1	*salivarius*	*parasalivarius*	99.4	1.46	92	2 228 781	FDR	42	F	34	Tooth	–	11	0	27.02 (20)
JACLQN000000000	16.1	*salivarius*	*parasalivarius*	99.4	1.46	79	2 216 454	FDR				Throat				
JACLQM000000000	17.1	*salivarius*	*parasalivarius*	99.4	1.46	60	2 222 428	FDR								
JACLQL000000000	21.1	*salivarius*	*parasalivarius*	99.68	0.22	34	2 114 746	RA	62	F	25	Tooth, GI	MTX	49	976	1.62 (47)
JACLQK000000000	22.1	*salivarius*	*parasalivarius*	99.82	0.88	60	2 394 449	RA	53	F	32	Throat, UTI	MTX	36	23	4.3 (10)
JACLQJ000000000	23.2	*salivarius*	*parasalivarius*	99.69	0.58	44	2 160 453	RA	32	F	18	–	SSZ	110	83	1.68 (46)

Genome sequences were compared for isolates shaded green.
ACPA, anti-citrullinated peptide autoantibody; BMI, body mass index; F, female; FDR, first-degree relatives; GI, gastrointestinal; HC, healthy controls; M, male; MTX, methotrexate; RA, rheumatoid arthritis; RF, rheumatoid factor; SSZ, sulfasalazine; UTI, urinary tract infection.

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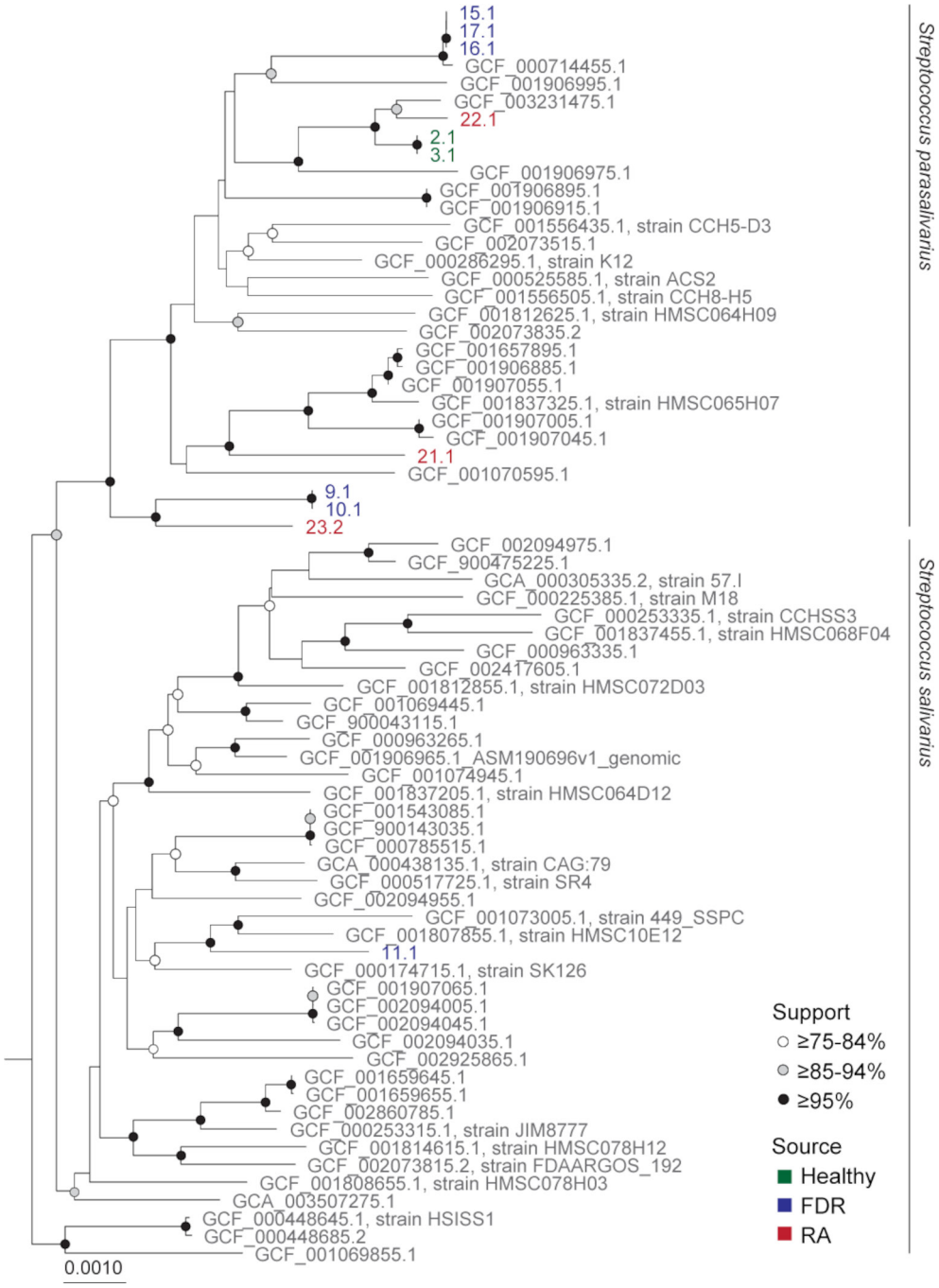


Figure 3 Phylogenetic analysis of *S. parasalivarius* and *S. salivarius* isolates. Maximum-likelihood tree generated using IQ-TREE from alignment of 120 marker genes identified using GTDB-Tk. Alignment filtered for sites conserved in a minimum of 10% of genomes yielding 39 247 residues. IQ-TREE built-in model testing assigned model JTT+F+R5 to data set. Bootstrap support values from IQ-TREE Ultrafast Bootstrapping with 1000 replicates. Isolate genomes marked in red (RA), blue (FDR) and green (healthy). Instances of multiple isolates from the same individual are clustered: 2.1/3.1, 9.1/10.1 and 15.1/16.1/17.1. Public genomes from NCBI representing *S. salivarius* and *S. parasalivarius* included as defined by GTDB R04-RS89 (grey). *S. vestibularis* genome GCF_000188295.1 used as outgroup. FDR, first-degree relatives; RA, rheumatoid arthritis.

supplemental table 10). Multiple encoded proteins displayed a pattern of presence/absence matching the TNF profile, including several potentially involved in cell wall synthesis, such as the FemA/B homologue, which contributes to peptidoglycan bridge formation and several glycosyltransferase proteins—of interest given the known immunogenicity of SCW (online supplemental tables 11 and 12).

Untreated SKG mice are healthy, and 100% of female SKG mice develop arthritis after one intraperitoneal injection of microbial beta-glucan (curdlan).[41] To assess potential for SCW of the different isolates to induce disease, we generated SCW from RA 21.1 and 23.2, and HC 2.1. We compared their arthritogenicity after two intraperitoneal injections, to standard *S. pyogenes* SCW, 10S PS-PG in female SKG mice. While the time course of response differed somewhat between experiments, all mice developed arthritis after 10S PS-PG or study SCW (figure 4D and E). Joint swelling scores were not significantly different between mice receiving the study SCW, and disease was generally more severe after 10S PS-PG. Curdlan-induced arthritis developed more rapidly and was more severe than SCW arthritis

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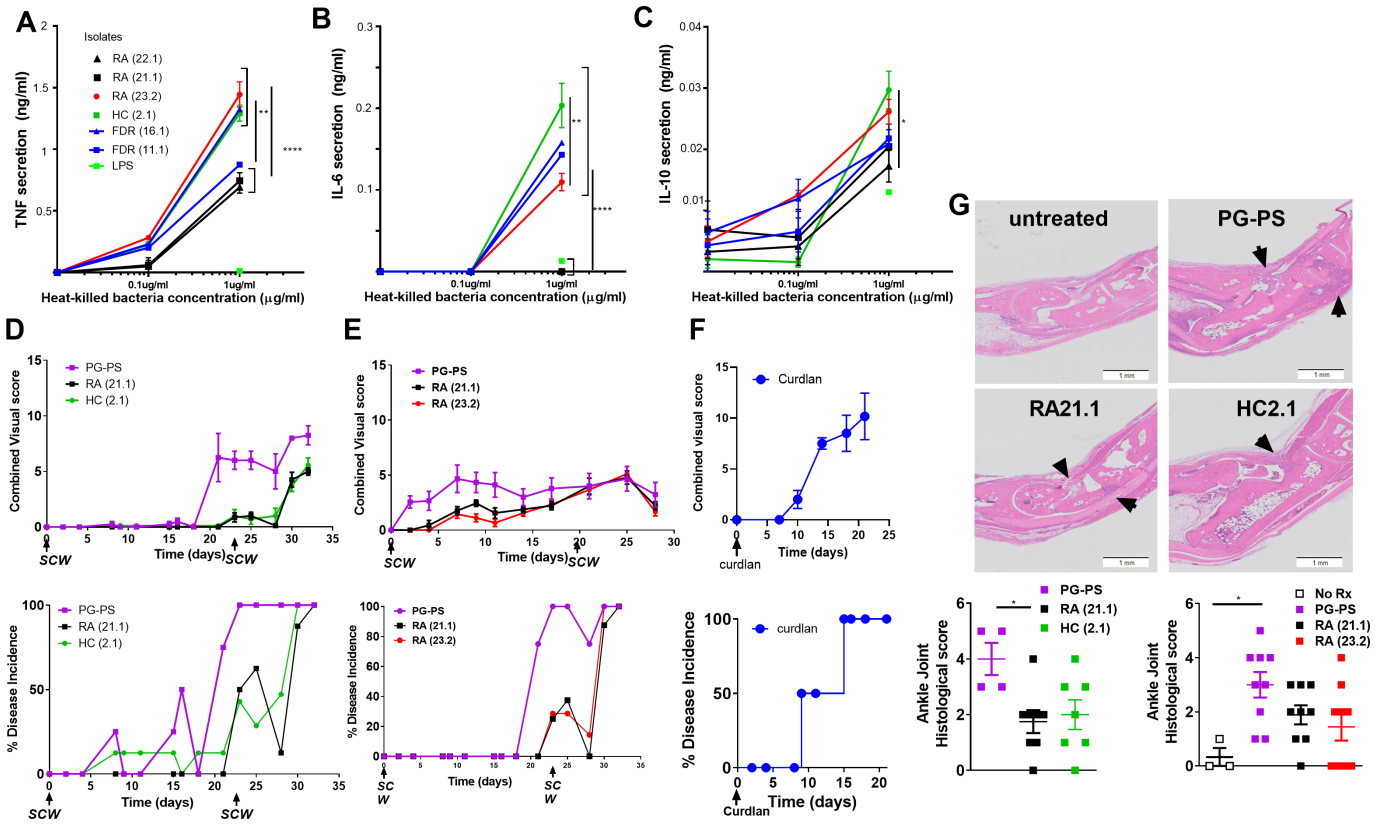


Figure 4 Functional characterisation of streptococcal strains in vitro and in vivo varying concentrations of heat-killed *S. parasalivarius* (2.1, 21.1, 22.1, 23.2, 16.1) and *S. salivarius* (11.1) isolates were incubated with SKG splenocytes for 24 hours then TNF (A), IL-6 (B) and IL-10 (C) were measured in cell free supernatant by cytokine bead array. 10S PG-PS or SCW from Streptococcal isolates 21.1, 2.1 () or 21.1 and 23.2 (E) were administered intraperitoneally to female SKG mice then visual score of ankle and wrists, and arthritis incidence, were monitored three times weekly for 32 () or 28 (E) days. Curdlan was administered intraperitoneally and visual score and arthritis incidence were monitored for 21 days (F). Joint histology was scored at the endpoint of each experiment (). Arrowheads: mild arthritis in the ankle and small joints of foot, arrows: moderate ankle and foot arthritis and soft tissue inflammation. FDR, first-degree relatives; HC, healthy controls; IL, interleukin; LPS, lipopolysaccharide; PG-PS, peptidoglycan-polysaccharide; RA, rheumatoid arthritis; SCW, streptococcal cell wall;TNF, tumour necrosis factor.

(figure 4F). Joint swelling scores were not significantly different between mice receiving the study SCW and disease was generally more severe after 10S PS-PG. Histologically, whereas mice receiving study SCW developed mild arthritis in the small joints of the feet and ankle, 10S PS-PG induced more florid ankle, foot and soft tissue inflammation (figure 4G). Thus, similar to TNF-dependent Chlamydia-induced arthritis in SKG mice,[42] repeated in vivo exposure to SCW is sufficient to trigger mild-to-moderate chronic inflammatory arthritis in this autoimmune-prone host, regardless of the disease-association status of the oral community from which the streptococci derived.

DISCUSSION

Using 16S rRNA gene amplicon sequencing, we show that patients with RA and some FDR, who were more likely to report bleeding gums than age and sex-matched HC, had dysbiotic oral microbiome profiles. Oral dysbiosis in FDR—with unknown disease class—could be classified according to similarity to the RA oral dysbiosis score. The dysbiosis score was not associated with clinical features surveyed among participants—including HLA-DR type, autoantibody titre, history of infection, medications, disease duration, joint count or erythrocyte sedimentation rate/C-reactive protein inflammatory markers in patients with RA or FDR. However, we did not formally measure disability or damage, and no longitudinal data were obtained to assess impact of disease activity on dysbiosis. Our study examined tongue swabs, as the tongue has the most stable, diverse and dense bacterial biofilm of the oral cavity.[35] Tongue microbiota are comparable to that of saliva, of clinical relevance due to the association of oral species with the gut environment in RA.[8] Since this compartment differs from the gingival or supragingival plaque biofilm, it is not surprising that we did not identify differential abundance of periodontitis keystone bacteria. Most of the patients with RA were treated with conventional or biological DMARDs, which may also have influenced the results. Furthermore, in absence of data from a second RA cohort, we could not fully validate the oral dysbiosis score. Nevertheless, our data support previous observations of dysbiosis of gut and oral microbiomes in RA.[5]

The 16S rRNA gene amplicon sequencing revealed five *Streptococcus* OTUs that were differentially abundant between RA and HC samples. With culture-based isolation and whole genome sequencing, we recovered 15 streptococcal isolates across three different species, *S. parasalivarius* (n=10), *S. salivarius* (n=1) and *S. parasanguinis_B* (n=4), with strains of *S. parasalivarius* isolated from RA, FDR and HC subjects. While only *S. parasalivarius* was isolated from patients with RA, culturing efforts were not exhaustive, based on comparison of the genomic data with the amplicon sequencing data (online supplemental table 6). Phylogenetic comparison of the *S. parasalivarius* isolates revealed significant genomic diversity across the RA-associated strains. Exposure of splenocytes to heat-killed *S. parasalivarius*

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isolates in vitro differentially promoted the production of innate immune cytokines IL-6 and TNF, potentially reflecting the observed genomic diversity. Importantly though, all strains dominantly induced TNF secretion, and TNF is likely to contribute to their in vivo pathogenicity in SKG mice.[42]

PAMPs associated with multiple pathobionts expanding in the oral dysbiotic environments could influence innate immune control of the mucosal barrier, such as the gingiva, predisposing to or exacerbating periodontitis. In hypervascular inflammatory tissue, permeability to resident mucosal microbes increases, enhancing systemic haematogenous spread of microbes, debris or PAMPs. Ectopic intestinal colonisation by oral bacteria, including streptococcal species, may also elicit intestinal inflammation.[43] Multiple encoded proteins within the *S. parasalivarius* isolates could potentially contribute to their immune-stimulatory function, including several involved in SCW synthesis. SCW consist of peptidoglycan, which induces inflammation, and polysaccharide, which protects against enzyme degradation in vivo.[32] The barrier that polysaccharides present to streptococcal phagocytosis by macrophages necessitates antibody and complement opsonisation for clearance. After intraperitoneal injection of SCW, female Lewis rats were shown to develop a chronic inflammatory arthritis with synovial localisation of the degradation-resistant SCW. Here they were phagocytised, but with persistence of PAMPs within the inflammatory site.[11] Depletion of macrophages with clodronate liposomes suppressed arthritis development.[44] By contrast, in BALB/c mice, SCW acute arthritis resolves spontaneously, and continued joint injections induce chronic arthritis.[33] IL-6, IL-17 and IL-23 are the predominant cytokines released in SCW-induced inflammatory arthritis.[45] SKG mice (BALB/c background) are genetically prone to Th17 or TNF-mediated chronic inflammatory arthritis after exposure to microbial products or chlamydial infection, due to reduced bacterial control by the host immune system.[46] In the present study, two intraperitoneal administrations of SCW derived from *S. parasalivarius* isolates induced mild chronic inflammatory arthritis in SKG mice, demonstrating that the ZAP-70 mutation prevents resolution of SCW arthritis. Furthermore, systemic administration was sufficient for development of arthritis, suggesting focal haematogenous spread of SCW debris.

Together, our data suggest that local inflammation and barrier dysfunction in the gingiva, permitting pro-inflammatory SCW to penetrate and reach the joint, are more important than the source of *Streptococcus* species from which SCW are derived. Controlling mucosal inflammation and barrier dysfunction in genetically predisposed individuals with oral dysbiosis and local periodontal inflammation—associated with poor dental health, smoking and poor nutrition—should reduce their propensity for arthritis development.

Description of *Streptococcus parasalivarius* sp nov

Streptococcus parasalivarius (pa.ra.sal'i.var'i.us. Gr. prep. *para* resembling; N.L. part. adj. *salivarius* specific epithet of *Streptococcus salivarius*; N.L. part. adj. *parasalivarius* (*Streptococcus*) *salivarius*-like). Represented by isolate genome 21.1 (acc. no. JACLQL000000000) obtained from a tongue swab of a patient with RA.

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Competing interests None declared.

Patient and public involvement statement At what stage in the research process were patients/the public first involved in the research and how? Patients and the public were involved at the recruitment stage. How were the research question(s) and outcome measures developed and informed by their priorities, experience and preferences? No direct patient involvement. How were patients/the public involved in the design of this study? Not involved. How were they involved in the recruitment to and conduct of the study? Not involved. Were they asked to assess the burden of the intervention and time required to participate in the research? No. How were (or will) they be involved in your plans to disseminate the study results to participants and relevant wider patient communities (eg, by choosing what information/results to share, when and in what format)? Preliminary results were shared via a newsletter to the involved participants to inform them of study progress.

Patient consent for publication Not required.

Ethics approval Studies approved by Metro South Hospital and Health Service (HREC/94/QPAH/6), The University of Queensland (2012000275) and by Greenslopes Private Hospital (GREC 14/53) Human Research Ethics Committees.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement Data are available in a public, open access repository. All data relevant to the study are included in the article or uploaded as supplementary information. Assembled genomes and raw 16S rRNA gene amplicon sequencing data are available via NCBI BioProject PRJNA656387. Prokka annotated genomes available at https://github.com/katebowerman/Streptococcus. Other data are available on request to corresponding author.

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